

Tunable Infrared Sensing Properties of MXenes Enabled by Intercalants

Jun Tang, Hujie Wan, Libo Chang, Bihua Hu, Shuyu Cui, Yulong Chen, Wei Chen, Junjie Hao, Haodong Tang, Xingzhu Wang, Kai Wang,* Chuanfang (John) Zhang,* Qiye Wen, Xu Xiao,* and Baomin Xu*

Light interaction physical properties of MXenes are fundamentally important and promising for developing new broadband photodetection applications. In this work, the significant influence of intercalants on the photo-electronic properties of MXenes is studied. It is found that the intercalated H₂O-induced inter-flake resistance greatly restricts the photon excitation responsivity of Ti₃C₂T_x, resulting in a slow and irreversible resistance change under laser irradiation. After deintercalation of H₂O by in situ laser writing, the resistance of Ti₃C₂T_x responds fast and reversibly to laser excitation and the variation range is proportional to laser power. By extending the findings to another MXene member, Mo₂TiC₂, the intercalant induced resistance is successfully developed for optoelectronic detection. By selectively deintercalating H₂O in the as-prepared Mo₂TiC₂ which is co-intercalated by H₂O and TMAOH, bare TMAOH intercalated Mo₂TiC₂ shows fast and reversible resistance changes with laser. Moreover, the resistance of bare TMAOH intercalated Mo₂TiC₂ shows a negative temperature coefficient behavior while the fully deintercalated Ti₃C₂T_x shows a positive temperature behavior. These findings elucidate the understanding of the optoelectronic properties of MXenes and provide insights into the engineering of interfacial chemistry of MXenes for performance enhancement toward various detectors/sensors beyond optoelectronic applications.

1. Introduction

Transition metal carbides, nitrides and carbonitrides, known as MXenes, are a rapidly emerging class of two-dimensional (2D) materials with diverse applications in energy storage, electromagnetic interference (EMI) shielding, optoelectronic devices, environmental and water treatment, sensors and biochemical applications.^[1–20] Titanium carbide MXene (Ti₃C₂T_x, where T represents the surface terminations such as -F, -OH and =O), as the first reported MXene in 2011, has been well studied and exhibits unique electronic properties, electrochemical properties and surface chemistry varieties.^[21–26] The conductivity of Ti₃C₂T_x, in particular, largely determines their unique applications.^[4,11,27] For example, higher conductivity of Ti₃C₂T_x is desired in EMI shielding while rapid changes in conductivity are preferred in sensors and optoelectronic detectors, respectively.^[22,28] Recent works intensively reported on developing the

J. Tang, B. Hu, S. Cui, Y. Chen, X. Wang, B. Xu
Department of Materials Science and Engineering
Southern University of Science and Technology
Shenzhen, Guangdong Province 518055, P. R. China
E-mail: xu.bm@sustech.edu.cn

H. Wan, L. Chang, Q. Wen, X. Xiao
State Key Laboratory of Electronic Thin Film and Integrated Devices
School of Electronic Science and Engineering (Chengdu) and Yangtze
Delta Region Institute (Huzhou)
University of Electronic Science and Technology of China
Chengdu, Sichuan Province 610054, P. R. China
E-mail: xuxiao@uestc.edu.cn

W. Chen, J. Hao, H. Tang, K. Wang
Department of Electronic Engineering
Southern University of Science and Technology
Shenzhen, Guangdong Province 518055, P. R. China
E-mail: wangk@sustech.edu.cn

W. Chen
College of Engineering Physics
Shenzhen Technology University (SZTU)
Lantian Road 3002, Shenzhen, Pingshan 518118, P. R. China

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/adom.202200623>.

J. Hao, H. Tang, K. Wang, B. Xu
Guangdong-Hong Kong-Macao Joint Laboratory for
Photonic-Thermal-Electrical Energy Materials and Devices
Shenzhen, Guangdong Province 518055, P. R. China

X. Wang
Academy for Advanced Interdisciplinary Studies
Southern University of Science and Technology
Shenzhen, Guangdong Province 518055, P. R. China

X. Wang
Shenzhen Putai Technology Co., Ltd
Shenzhen 518110, P. R. China

C. (J.) Zhang
College of Materials Science & Engineering
Sichuan University
Chengdu, Sichuan 610065, P. R. China
E-mail: chuanfang.zhang@empa.ch

C. (J.) Zhang
Swiss Federal Laboratories for Materials Science and Technology (Empa)
ETH Domain
Überlandstrasse 129, Dübendorf CH-8600, Switzerland

DOI: 10.1002/adom.202200623

optoelectronic applications of MXenes due to their tunable electronic structures, high electronic conductivity, and high compatibility with mainstream semiconductor processes.^[29–37] Among these applications, the as measured resistance variations of acid-etched MXenes with plenty of aqueous intercalants had been widely used directly as spontaneous signals to stimulations. Nevertheless, the measured resistance is in fact of a coupled resistance of MXenes which contains two disparate physical meanings behind: intra-flake resistance (R_{intra}) which reflects the intrinsic metallic conductivity of MXenes, and inter-flake resistance (R_{inter}) which relates to the surface terminations and intercalants.^[38] The two resistances react specifically to external stimuli, resulting in severely limited responsive qualities. For example, in thermal sensor applications, metallic R_{intra} increases with increasing the temperature, while R_{inter} decreases since intercalated H_2O molecules between MXene nanosheets are thermally evacuated. As a result, the measured resistance of MXenes cannot be simply used as the signal for precise sensing. Consequently, the coupled resistances of MXenes may hinder the development of reliable optoelectronic applications, where the resistance variations can be intensively induced by the photothermal effect in $\text{Ti}_3\text{C}_2\text{T}_x$ films and lead to inaccurate results for photon-electron responsivity.

Therefore, decoupling R_{inter} and R_{intra} is necessary to further optimize the sensing/detecting performance of MXenes on specific sensing/detecting applications. In particular, R_{inter} is dominating for sensing/detecting changes in many environmental circumstances such as gases, humidity, lighting, and temperature since the intercalants in MXenes can be easily influenced by these stimuli. And R_{intra} , related to the metallic property of MXenes, is sensitive to detect both temperature variations and optoelectronic detections by considering the high photothermal conversion efficiency of MXenes.^[39–47] To obtain an accurate detecting signal, R_{inter} and R_{intra} need to be decoupled in real detecting applications.^[44] However, in most of the reported works, MXene colloids were directly used for preparing thin films for optoelectronic applications, where numerous intercalants from the colloids remain in MXene films. Very limited work reported so far on the mechanism of decoupling of R_{inter} and R_{intra} in MXenes, neither pay any attention to MXene sensing/detecting performance based on decoupled resistances, which significantly hinders the development of MXenes based optoelectronic applications. A better understanding of the optoelectronic properties of MXenes is not only important for performance improvements of MXene based detectors but also helpful in explaining abnormal phenomena such as rapid responsivity fading with laser powers.^[29]

In this work, we reported an interesting phenomenon that the resistance changes of as-prepared $\text{Ti}_3\text{C}_2\text{T}_x$ films with H_2O intercalants are slow and irreversible with near-infrared (NIR) laser irradiation. By methodically eliminating H_2O in $\text{Ti}_3\text{C}_2\text{T}_x$ interlayers, the influence of intercalated H_2O molecules on the optoelectronic response was minimized and the resistance of deintercalated $\text{Ti}_3\text{C}_2\text{T}_x$ was able to respond fast and reversibly. After complete deintercalation, the change in R_{intra} dominates the overall resistance variations of $\text{Ti}_3\text{C}_2\text{T}_x$ film, whereas R_{inter} remains unchanged under laser irradiation. As a result, the overall resistance of $\text{Ti}_3\text{C}_2\text{T}_x$ film exhibits a positive temperature coefficient (PTC) behavior since R_{intra} is proportional to

temperature. On the contrary, tetramethylammonium hydroxide (TMAOH) intercalated $\text{Mo}_2\text{TiC}_2\text{T}_x$ exhibits a negative temperature coefficient (NTC) behavior. Because R_{intra} of $\text{Mo}_2\text{TiC}_2\text{T}_x$ is generally temperature inert, R_{inter} , which is reversely proportional to temperature and will dominate the overall resistance variation. The findings in this work unveil a fundamental understanding of the independent effects of R_{inter} and R_{intra} on the sensing/detecting performances of MXenes and shed light on the engineering of MXene interlayer chemistry for effective performance improvement.

2. Results and Discussions

2.1. Reversible Intercalation–Deintercalation of H_2O in $\text{Ti}_3\text{C}_2\text{T}_x$

$\text{Ti}_3\text{C}_2\text{T}_x$ is generally synthesized by selectively etching Al atoms from its precursor Ti_3AlC_2 , followed by delamination to obtain predominantly single-layer $\text{Ti}_3\text{C}_2\text{T}_x$ nanosheets, forming aqueous colloids (Figure S1, Supporting Information).^[48] The as-prepared $\text{Ti}_3\text{C}_2\text{T}_x$ colloids can then be sprayed or filtrated into films (Figure S1, Supporting Information and Figure 1a), which can subsequently be laser patterned into line-shaped devices as shown in Figure 1b. It's worth noting that even after vacuum drying, there is still plenty of intercalated H_2O in the interlayer spacing of $\text{Ti}_3\text{C}_2\text{T}_x$ films. To directly assess the influence of intercalated H_2O , the resistance variations of the $\text{Ti}_3\text{C}_2\text{T}_x$ line-shaped device are analyzed in dry Ar flow and moist Ar flow, the results of which are presented in Figure 1c. In general, the resistance of $\text{Ti}_3\text{C}_2\text{T}_x$ continuously decreases in dry Ar flow to $\approx 72\%$ of its initial value until the resistance gets stabilized. When dry Ar flow is turned off and moist Ar flow is switched on, the resistance of $\text{Ti}_3\text{C}_2\text{T}_x$ increases. As demonstrated in Figure 1d, the intercalated or adsorbed H_2O in $\text{Ti}_3\text{C}_2\text{T}_x$ can be partly blown away in Ar flow, resulting in a drop in inter-flake resistance (R_{inter}). Excess H_2O molecules can reconstruct $\text{Ti}_3\text{C}_2\text{T}_x$ film when moist Ar is introduced, resulting in a rise in R_{inter} . The resistance of $\text{Ti}_3\text{C}_2\text{T}_x$ becomes steady after 100 s in moist Ar, which can be reduced again by turning off moist Ar flow and switching on dry Ar flow again. These results reveal that H_2O intercalation/deintercalation process is highly reversible and the resistance of $\text{Ti}_3\text{C}_2\text{T}_x$ is associated with the intercalants.

2.2. Influences of Intercalated H_2O on the Electronic Conductivity of $\text{Ti}_3\text{C}_2\text{T}_x$ Under NIR Laser Irradiation

As reported, MXenes exhibit ultrahigh photothermal conversion efficiency up to 100%.^[31] Accordingly, the temperature of $\text{Ti}_3\text{C}_2\text{T}_x$ film will increase when irradiated by an NIR laser, resulting in a direct increase in the metallic resistance of $\text{Ti}_3\text{C}_2\text{T}_x$ (R_{intra}). In this work, the temperature elevation is modulated by the laser power as shown in Figure S2 (Supporting Information). However, when the NIR laser (the scanning mode of the laser is displayed in Figure S3, Supporting Information) was applied to the spray-coating $\text{Ti}_3\text{C}_2\text{T}_x$ film, an abnormal decrease in resistance was observed, and the resistance slowly recovers after laser stimulation (Figure 2a). Even though R_{intra} increases in

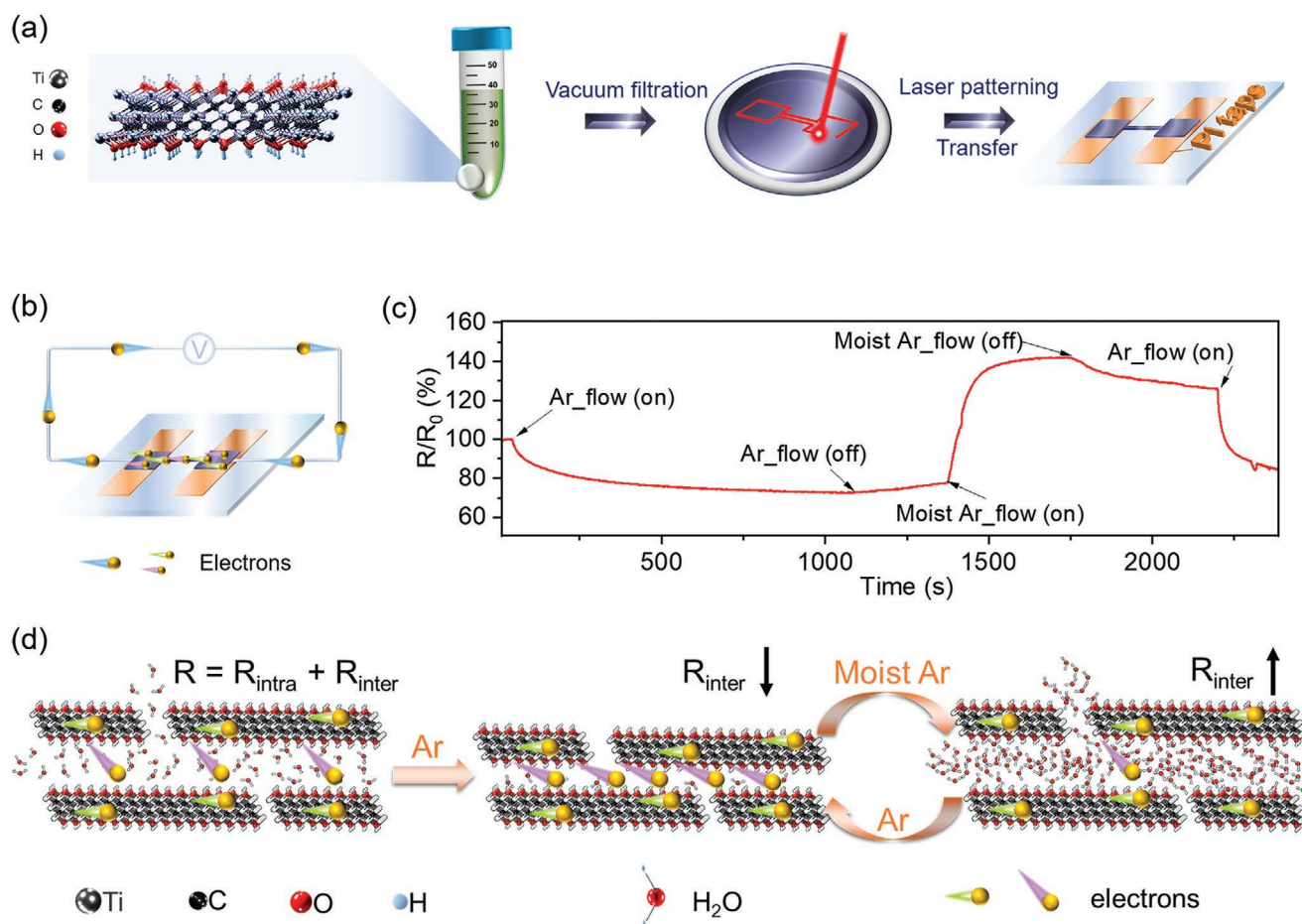


Figure 1. Fabrication process of the freestanding Ti₃C₂T_x line-shaped device and its resistance variations induced by intercalated H₂O. a) Schematic of the fabrication process of freestanding Ti₃C₂T_x line-shaped device from its aqueous colloids. The as-prepared Ti₃C₂T_x colloid was firstly filtrated into freestanding films, which were then transferred onto a glass slide with both ends fixed by a PI tape. The suspended middle part of Ti₃C₂T_x film was then laser patterned into a line with a width of 0.5 mm. b) The line-shaped device was connected to a CHI660 electrochemical workstation to in situ detect the resistance changes of Ti₃C₂T_x. c) The resistance variations and d) schematic illustration of Ti₃C₂T_x film in different atmospheres.

irradiation, the laser-induced temperature elevation inevitably causes the deintercalation of H₂O, resulting in a reduction of R_{inter} . This phenomenon indicates that the changes in overall resistance R of Ti₃C₂T_x is modulated by both ΔR and ΔR_{intra} . In the case of spray-coating Ti₃C₂T_x film, ΔR_{inter} appears to dominate the overall resistance variation (ΔR), as there are no significant variation signals for R_{intra} which is proportional to temperature (Figure 2a). Since the deintercalation process is reversible as shown in Figure 1c, R_{inter} of the spray-coating Ti₃C₂T_x gradually recovers in the humid air after laser, which explains the recovery of the overall resistance as shown in Figure 2a. According to our ideal model, if Ti₃C₂T_x was fully deintercalated, R_{inter} remains constant with temperature increasing and thus ΔR_{intra} dominates the ΔR in turn. To verify this assumption, spray-coating Ti₃C₂T_x is placed in dry Ar flow condition to remove the adsorbed/intercalated H₂O (Figure S4a, Supporting Information). In contrast to the results obtained in the air (Figure 2a), Ti₃C₂T_x in Ar exhibits a positive temperature coefficient of resistance in laser cycles as shown in Figure S4b (Supporting Information), which reflects the typical changes of R_{intra} with laser. In the following 100 laser cycles in Ar, the

baseline of R continuously decreases gradually toward stabilized tendency as shown in Figure S4c (Supporting Information), which implies a maximum removal of intercalated H₂O by laser. Though the overall resistance exhibited decreasing in overall tendency, the amplitude of ΔR in each laser on/off cycle almost remains constant, which suggests that the R_{intra} is almost intercalant-irrelevant and can reliably reflect the power density of the irradiated laser. Briefly, by comparing the results in the air and Ar, we can conclude that laser-induced ΔR is dominated by ΔR_{inter} for as-prepared spray-coating Ti₃C₂T_x with H₂O as intercalants, which will be dominated by ΔR_{intra} by deintercalating H₂O. Further evidence is also found from the results of spray-coating Ti₃C₂T_x in consecutive laser cycles in the air as shown in Figure S5 (Supporting Information).

It is worth noting that the recovery of R_{intra} in spray-coating Ti₃C₂T_x is relatively slower than the elevation process in each laser cycle as shown in the insets of Figures S4c and S5c (Supporting Information). In situ real-time temperature mappings reveal that the cooling down process takes several seconds for spray-coating Ti₃C₂T_x (Figure S6a, Supporting Information), which is highly consistent with the time taken by the recovery

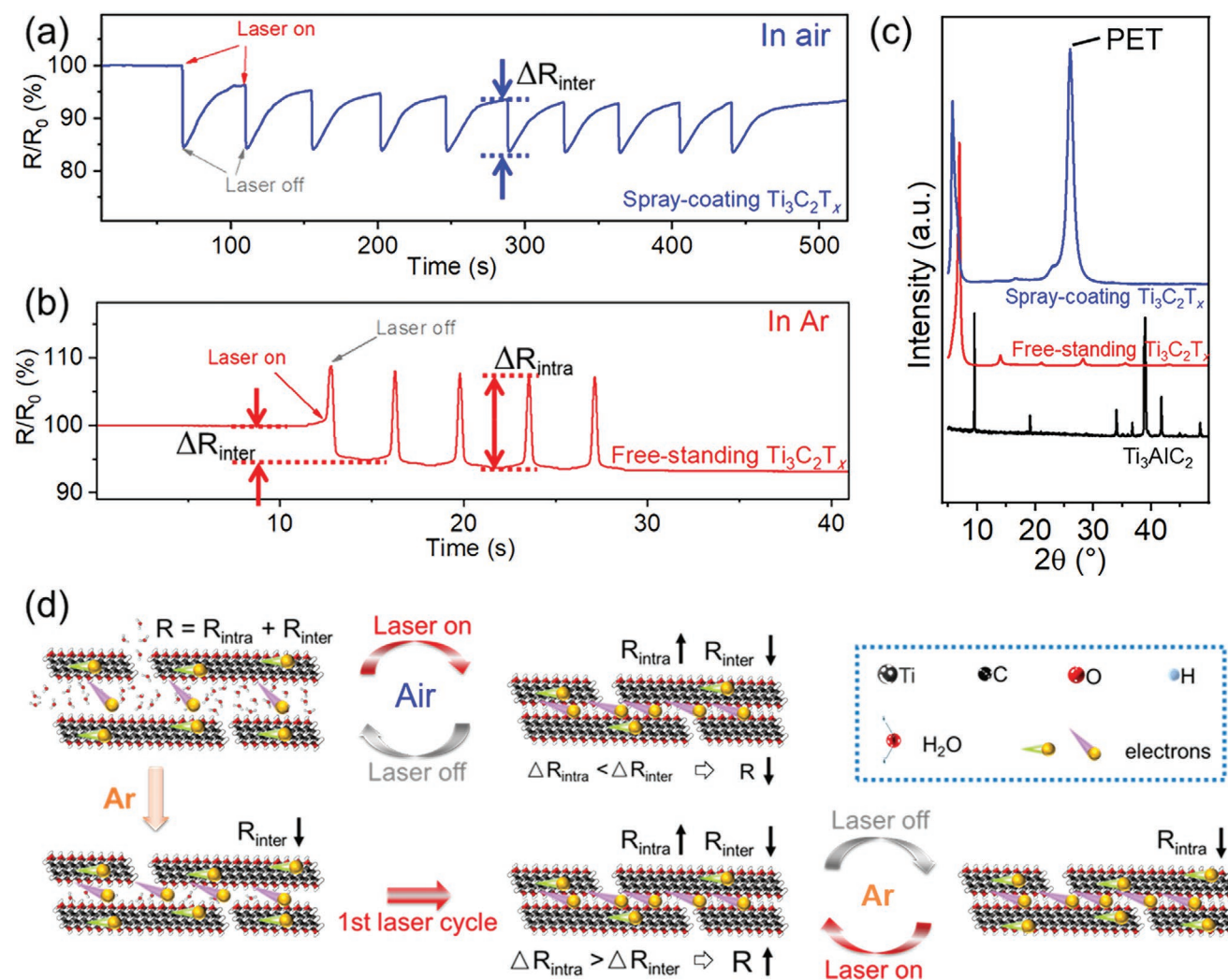


Figure 2. NIR laser detection performance of the spray-coating $\text{Ti}_3\text{C}_2\text{T}_x$ in air and freestanding $\text{Ti}_3\text{C}_2\text{T}_x$ in Ar. a) The resistance changes of pristine spray-coating $\text{Ti}_3\text{C}_2\text{T}_x$ in air and b) freestanding $\text{Ti}_3\text{C}_2\text{T}_x$ in Ar flow in laser on/off cycles, respectively. c) XRD patterns and d) schematic illustrations of the resistance changes of $\text{Ti}_3\text{C}_2\text{T}_x$ induced by laser in the air and Ar. The laser power was set to be 6% of the maximum value, the practical output power density of which was estimated to be $\approx 2.64 \text{ W mm}^{-2}$.

of resistance (inset of Figure S4c, Supporting Information). To avoid the influence of substrates, a suspended freestanding $\text{Ti}_3\text{C}_2\text{T}_x$ line-shaped device is designed and fabricated as shown in Figure 1a,b, which shows a fast cooling down process within only 1 s (Figure S6b, Supporting Information). This could be ascribed to the differences between the alignments of $\text{Ti}_3\text{C}_2\text{T}_x$ flakes in the two films. According to the X-ray diffraction (XRD) results shown in Figure 2c, the interlayer spacing of freestanding $\text{Ti}_3\text{C}_2\text{T}_x$ exhibits a larger diffraction angle of (002) peak which means a relatively smaller inter-flake distance in the thin-film. As a result, the freestanding film with a more compact flake alignment demonstrates enhanced thermal conduction. Also, the cooling down process of spray-coating $\text{Ti}_3\text{C}_2\text{T}_x$ films could be greatly affected by the low thermal conductivity of the plastic PET substrate. Accordingly, the resistance of freestanding $\text{Ti}_3\text{C}_2\text{T}_x$ responds much faster in the laser on/off cycles than the spray-coating one (Figure 2b and Figure S8, Supporting Information).

2.3. Excellent Responsive Conductivity Changes of Deintercalated $\text{Ti}_3\text{C}_2\text{T}_x$ with NIR Laser

Since freestanding $\text{Ti}_3\text{C}_2\text{T}_x$ films exhibit fast and reversible resistance changes to infrared laser, we further investigated their responsivity under laser with different power densities (Figure 3a). The power-dependent resistance changes of a freestanding $\text{Ti}_3\text{C}_2\text{T}_x$ line-shaped device are initially investigated by gradually tuning the laser power from 2.64 to 6.04 W mm^{-2} and finally back to 2.64 W mm^{-2} (Figure 3b), followed by 1000 laser on/off cycles with laser power of 2.64 W mm^{-2} . After cycling, the relationship between resistance change and laser power is assessed again (Figure 3c). Conclusively, the amplitude of resistance variation in freestanding $\text{Ti}_3\text{C}_2\text{T}_x$ arises/reduces with the increase/decrease of laser power and keeps almost unchanged in the 1000 laser cycles. The quantitative results are shown in Figure 3d, revealing that the responsivity ($\Delta R/R_0$) is not reversible in the first power elevation–reduction process,

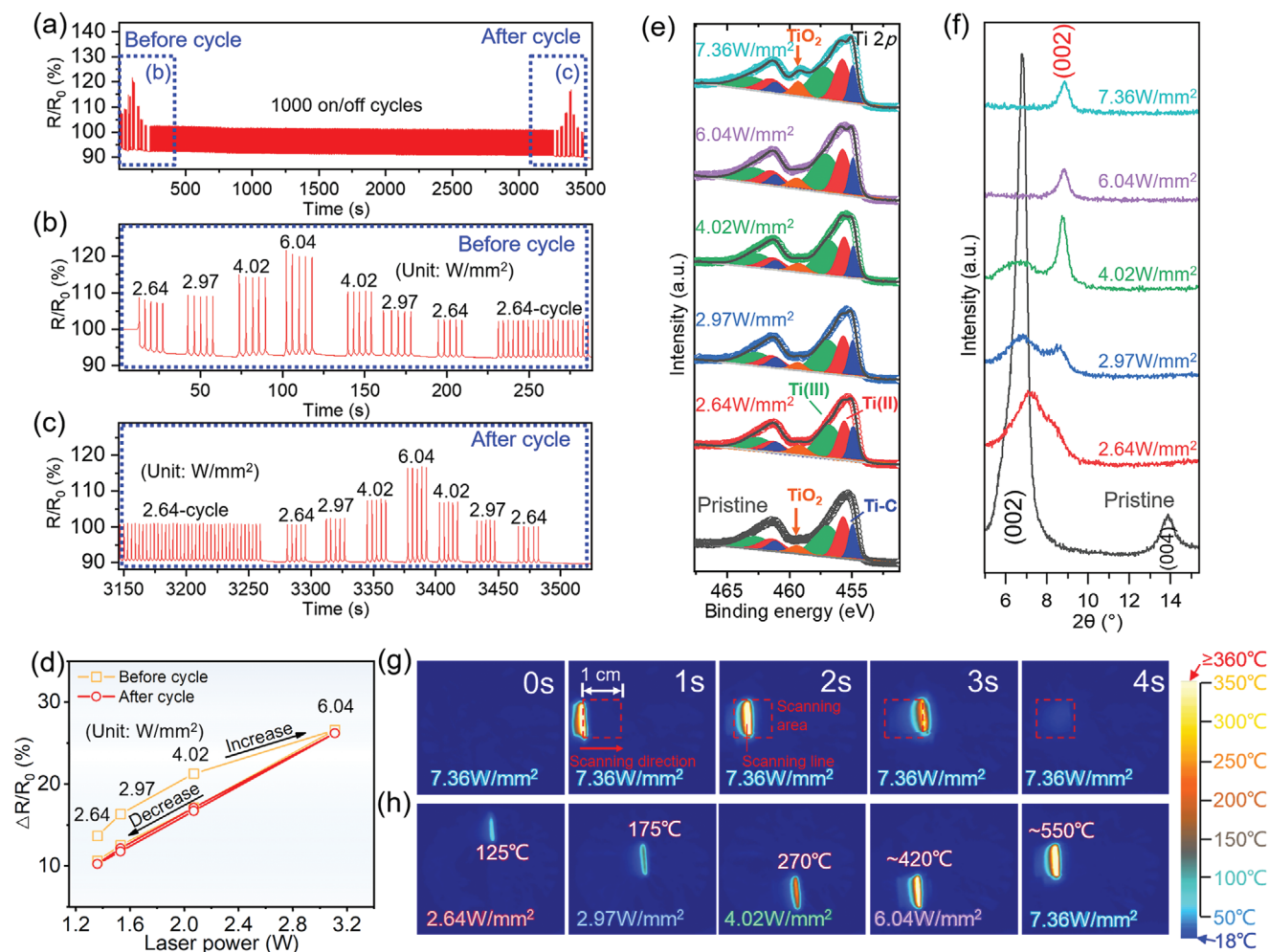


Figure 3. Resistance variations of freestanding $\text{Ti}_3\text{C}_2\text{T}_x$ and its structural and temperature changes induced by laser with different power. a) In situ resistance changes of freestanding $\text{Ti}_3\text{C}_2\text{T}_x$ line-shaped device induced by different lasers. The device was firstly tested in laser on/off cycles with different powers and then cycled for 1000 times with a constant power of 2.64 W mm^{-2} . After cycle, the device was cycled again in different power lasers. b) An amplified view of the resistance changes under different power lasers before and c) after cycles. d) The amplitude variation of ΔR with laser power before and after cycle. e) XPS results and f) XRD patterns of freestanding $\text{Ti}_3\text{C}_2\text{T}_x$ film treated by different power lasers. g) The photothermal temperature mappings of $\text{Ti}_3\text{C}_2\text{T}_x$ films induced by 7.36 W mm^{-2} laser at different scanning time. h) The photothermal temperature mappings of $\text{Ti}_3\text{C}_2\text{T}_x$ films induced by laser with different powers at the scanning time of 2 s.

which is ascribed to the deeper removal of intercalated H_2O under higher power densities. The overlapped part of the red and orange lines in Figure 3d indicates that the responsivity ($\Delta R/R_0$) in various power densities after 1000 cycles remains constant as the one in the first power-decreasing process, which indicates great cycling stability and reliability of freestanding $\text{Ti}_3\text{C}_2\text{T}_x$ on sensing NIR laser. The upper power limit of freestanding $\text{Ti}_3\text{C}_2\text{T}_x$ for sensing infrared laser is $\approx 12\%$ (corresponds to 6.04 W mm^{-2} , Table S1, Supporting Information), since a typical TiO_2 peak is observed in the XPS spectra of freestanding $\text{Ti}_3\text{C}_2\text{T}_x$ treated by 7.36 W mm^{-2} laser (Figure 3e). With laser powers lower than 6.04 W mm^{-2} , no oxidation takes place in $\text{Ti}_3\text{C}_2\text{T}_x$, while the interlayer spacing decreases as shown in Figure 3f. The (002) peak at 6.8° is assigned to the H_2O intercalated $\text{Ti}_3\text{C}_2\text{T}_x$, and the peak at 8.8° is assigned to the deintercalated $\text{Ti}_3\text{C}_2\text{T}_x$. The (002) peak at 6.8° (d -spacing = 1.3 nm) for pristine $\text{Ti}_3\text{C}_2\text{T}_x$ gradually decreases and a new (002) peak at 8.8°

(d -spacing = 1.0 nm) arises with an increase in laser power, indicating that interlayer H_2O is gradually removed the laser power elevating condition. As shown in Figure 3f, $\text{Ti}_3\text{C}_2\text{T}_x$ is not fully deintercalated until the laser power increases to 6.04 W mm^{-2} , which well explains the irreversibility of the resistance variation before the cycle (Figure 3b,d). Figure 3g,h shows the temperature mappings of 7.36 W mm^{-2} laser scanning line at a different time and laser lines at different powers, respectively. The quantitative temperatures of these lines are displayed in Figure S9 (Supporting Information). These results indicate that the resistance changes of deintercalated $\text{Ti}_3\text{C}_2\text{T}_x$ under laser may be ascribed to photothermally induced temperature variations. For example, the amplitude of the resistance variation ($\Delta R/R_0$) is $\approx 17\%$ under laser with a power of 4.02 W mm^{-2} , corresponding to a temperature elevation from 25 to $\approx 275^\circ\text{C}$. This amplitude is very close to the resistance change of $\text{Ti}_3\text{C}_2\text{T}_x$ observed in the same temperature range in a vacuum condition.^[30] Notably, the

deintercalated $\text{Ti}_3\text{C}_2\text{T}_x$ line-shaped device also works well in the air as shown in Figure S10 (Supporting Information) except for a reduced power window not higher than 4.02 W mm^{-2} since oxidation of $\text{Ti}_3\text{C}_2\text{T}_x$ takes place under this power density or higher as indicated by the XPS results (Figure S11, Supporting Information).

2.4. Influences of Intercalated TMAOH on the Electronic Conductivity of $\text{Ti}_3\text{C}_2\text{T}_x$ under NIR Laser Irradiation

We further studied the NIR sensing performance of H_2O and TMAOH co-intercalated $\text{Ti}_3\text{C}_2\text{T}_x$, the results of which are shown in Figure S12 (Supporting Information). The TMAOH is introduced in the synthesis process where TMAOH aqueous solution is used for delamination of multi-layered $\text{Ti}_3\text{C}_2\text{T}_x$ instead of LiCl. A drop in R_{inter} is observed after the first laser (2.64 W mm^{-2}) on/off cycle, which can be assigned to the removal of intercalated H_2O , similar to the results of H_2O - $\text{Ti}_3\text{C}_2\text{T}_x$ as discussed in Figure 2b. Another R_{inter} drop is observed when the laser power further increases to 4.02 W mm^{-2} , which may be caused by the decomposition of TMAOH. In general, the amplitude of ΔR in TMAOH- $\text{Ti}_3\text{C}_2\text{T}_x$ under laser (laser power $< 4.02 \text{ W mm}^{-2}$, Figure S12, Supporting Information) is relatively smaller than fully deintercalated $\text{Ti}_3\text{C}_2\text{T}_x$ (Figure 3b), which implies that TMAOH related R_{inter} may respond oppositely to R_{intra} of $\text{Ti}_3\text{C}_2\text{T}_x$ under laser irradiation condition, resulting in an impaired NIR sensing performance.

2.5. Influences of Intercalated H_2O and TMAOH on the Electronic Conductivity of Mo_2TiC_2 under NIR Laser Irradiation

On the other hand, if R_{intra} of MXene stays unchanged under laser irradiation, TMAOH related R_{inter} acts as an electric signal for NIR sensing showing a negative temperature coefficient behavior. Herein, Mo_2TiC_2 is chosen since its intra-flake resistance is almost temperature independent as reported by Hart et al.^[30] In this section, specifically, HF etched and TMAOH delaminated Mo_2TiC_2 is used for the assembly of freestanding films with both H_2O and TMAOH as intercalants. The resistance variations of Mo_2TiC_2 line-shaped device are then analyzed in Ar and moist Ar flow to evaluate the influence of intercalated H_2O , the results of which are presented in Figure 4a. It shows that the removal of intercalated or adsorbed H_2O in Ar flow results in a reduction of the resistance in Mo_2TiC_2 , which can be reverted when the device is placed in a moist Ar atmosphere. After blowing in Ar for ≈ 15 min, the resistance variations of Mo_2TiC_2 line devices in laser on/off cycles were investigated (Figure 4b). The resistance R of Mo_2TiC_2 shows a small drop after 2.64 and 2.97 W mm^{-2} laser cycles but significantly larger drops after 4.02 and 6.04 W mm^{-2} laser cycles (Figure 4b), which may be ascribed to the decomposition of TMAOH when the laser power increases to 4.02 W mm^{-2} as similar to TMAOH intercalated $\text{Ti}_3\text{C}_2\text{T}_x$.

XPS results confirm the decomposition of TMAOH as shown in Figure 4e,f. The Mo $3d$ spectra in Figure 4e show that Mo_2TiC_2 will not be oxidized until the laser power increases

to 7.36 W mm^{-2} . With laser power increasing to 4.02 W mm^{-2} or higher, the peak at 285.9 eV assigned to C–N bonds of TMAOH starts to decrease while the peak located at 284.9 eV of C–C bonds starts to increase (Figure 4e). When the laser power further increases to 7.36 W mm^{-2} , the C–N peak at 285.9 eV almost disappears, indicating a full decomposition of TMAOH. Notably, after 5 on/off cycles under a 7.36 W mm^{-2} laser irradiation, R of Mo_2TiC_2 remains consistent in lasers (Figure 4c), indicating intrinsic R_{intra} of Mo_2TiC_2 is NIR irrelevant. To develop the photodetection application of Mo_2TiC_2 , H_2O was selectively removed and TMAOH was retained by controlling the laser power since H_2O related R_{inter} responses slowly and irreversibly under laser as we have shown in Figure 2. Specifically, the Mo_2TiC_2 line-shaped device was treated by a 2.64 W mm^{-2} laser for 1000 cycles to step by step remove the intercalated H_2O . As shown in Figure 4b, H_2O -related R_{inter} continuously decreases, indicating that intercalated H_2O has been successfully removed step by step. Meanwhile, the amplitude of ΔR_{inter} (TMAOH) in each laser on/off cycle keeps almost unchanged, showing great cycling stability and reliability of TMAOH- Mo_2TiC_2 for sensing NIR laser. Accordingly, after deintercalation of H_2O , ΔR of TMAOH- Mo_2TiC_2 was able to be employed for indicating the laser power changes, showing a proportional relationship with the power density (Figure 4d).

Figure 4g shows a summary of the resistance changes of Mo_2TiC_2 in different conditions. In brief, the R of Mo_2TiC_2 can be divided into R_{intra} , $R_{\text{inter}}(\text{H}_2\text{O})$, and $R_{\text{inter}}(\text{TMAOH})$. $R_{\text{inter}}(\text{H}_2\text{O})$ of pristine Mo_2TiC_2 decreases with deintercalation of H_2O in Ar flow, the process of which can be reverted under moist Ar flow. After cycling in a low-power laser condition, the influence from $R_{\text{inter}}(\text{H}_2\text{O})$ on ΔR under laser has been greatly reduced and $\Delta R_{\text{inter}}(\text{TMAOH})$ dominates the change of the overall resistance. After a high-power laser stimulation, fully deintercalated Mo_2TiC_2 no longer shows any significant response in laser on/off cycles, since R_{intra} of Mo_2TiC_2 is temperature independent.

2.6. Summaries of Influences from Intercalants

Table 1 shows a summary of the response performance of different MXenes with various intercalants. In brief, the as-prepared MXenes are generally unsuitable for NIR detection. The intercalants in different MXenes should be carefully modified for improving the photodetection performance. In the case of Ti_3C_2 , all intercalants should be removed for NIR detection with a positive temperature coefficient, while in the case of Mo_2TiC_2 , interlayer TMAOH should be reserved for NIR detection with a negative temperature coefficient.

3. Conclusion

Understanding the roles of intercalants play in determining the optoelectronic properties of MXene films is essential for further improving the performance of photodetection and a better understanding of the abnormal optoelectronic phenomenon. In this work, we have systematically studied the impacts of intercalants on the NIR sensing performance of MXenes. Pristine

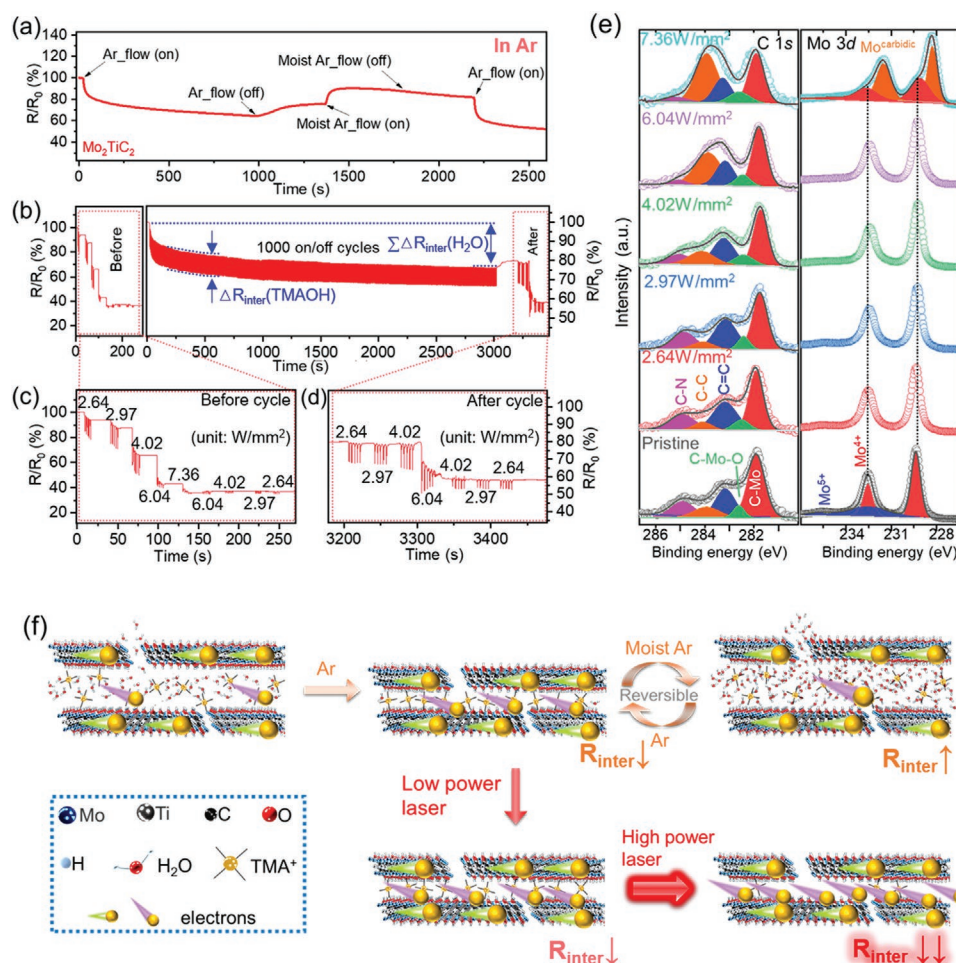


Figure 4. Resistance variations of freestanding Mo_2TiC_2 and its structural changes induced by laser with different power. a) The resistance variations of Mo_2TiC_2 film in different atmospheres. b) In situ resistance changes of freestanding Mo_2TiC_2 line-shaped device under laser irradiation. The device was firstly cycled in different power lasers and then cycled 1000 times with a 2.64 W mm^{-2} laser irradiation. After cycling, the device was cycled again in different power lasers. c) An amplified view of the resistance changes under different power lasers before and d) after cycles. e) C 1s and Mo 3d XPS spectra of Mo_2TiC_2 film treated by different power lasers. f) Schematic illustrations of the resistance changes of Mo_2TiC_2 in different circumstances and laser with different powers.

$\text{Ti}_3\text{C}_2\text{T}_x$ with H_2O as intercalant shows a slower recovery of R after laser irradiation, while deintercalated $\text{Ti}_3\text{C}_2\text{T}_x$ reveals faster R changes under laser stimulation. We showed that ΔR_{intra} of $\text{Ti}_3\text{C}_2\text{T}_x$ can be employed as signals for sensing NIR

laser with different power densities as long as ΔR_{inter} was suppressed by deintercalation. We further showed fully deintercalated Mo_2TiC_2 exhibits a stable R_{intra} under the laser condition, which is unsuitable for detecting NIR lasers. By selectively

Table 1. A summary of laser-induced resistance changes of MXenes with various intercalants.

MXenes	R_{inter}	R_{intra}	$R = R_{\text{inter}} + R_{\text{intra}}$	Dominating resistance change	IR sensing performance
	On Off	On Off	On Off		
$\text{H}_2\text{O-Ti}_3\text{C}_2$ (in air)	$\downarrow/\nearrow$	$\uparrow/\downarrow$	$\downarrow/\nearrow$	ΔR_{inter}	Bad
Deintercalated- Ti_3C_2 (in Ar)	$\rightarrow \rightarrow$	$\uparrow/\downarrow$	$\uparrow/\downarrow$	ΔR_{intra}	Good
TMAOH- Ti_3C_2 (in Ar)	$\downarrow/\uparrow$	$\uparrow/\downarrow$	$\uparrow/\downarrow$	ΔR_{intra}	Medium
Deintercalated - Mo_2TiC_2 (in Ar)	$\rightarrow \rightarrow$	$\rightarrow \rightarrow$	$\rightarrow \rightarrow$	–	Bad
TMAOH- Mo_2TiC_2 (in Ar)	$\downarrow/\uparrow$	$\rightarrow \rightarrow$	$\downarrow/\uparrow$	ΔR_{inter}	Good

Note: $\uparrow$ and $\downarrow$ represent that R increases and decreases fast, respectively. $\nearrow$ represents that R recovers slowly. $\rightarrow$ represents that R shows no significant changes under laser.

deintercalating H_2O , R_{inter} of TMAOH- Mo_2TiC_2 was able to be employed for NIR sensing, showing a fast response to the laser stimulation. It is worth noting that R_{intra} of $\text{Ti}_3\text{C}_2\text{T}_x$ and R_{inter} of TMAOH- Mo_2TiC_2 display a positive and a negative temperature coefficient behavior, respectively, showing different potentials in specified applications. In brief, by controlling the intercalants, we have successfully tuned the NIR sensing performance of different MXenes. These results shed light on designing and improving the optoelectronic performance of MXenes and MXene related materials/devices. In prospect, the optoelectronic performance of all the $\text{Ti}_3\text{C}_2\text{T}_x/\text{Mo}_2\text{TiC}_2$ based detectors may be significantly improved by selectively deintercalating interlayer H_2O in the light of the findings of this work.

4. Experimental Section

Synthesis of Pristine $\text{Ti}_3\text{C}_2\text{T}_x$ Suspensions: $\text{Ti}_3\text{C}_2\text{T}_x$ MXene was synthesized by selective etching of Al atoms from Ti_3AlC_2 (procured from Jilin 11 tech. Inc.) using HF/HCl etchant.^[27,34] Briefly, 12 mL of hydrochloric acid (HCl, technical grade, 35–38%), 2.5 mL hydrofluoric acid (Sigma Aldrich, 40%), and 6 mL of DI water were mixed by stirring. Then 1 g of Ti_3AlC_2 was slowly added to the solution at 35 °C followed by stirring at 400 rpm for 24 h. The as-obtained suspension (multi-layered MXene) was washed with deionized (DI) water via centrifugation at 5000 rpm (5 min per cycle) until pH $\geq$ 6. The sediment was collected and re-dispersed into 20 mL of water, and then added to the delamination solution containing 1 g LiCl and 40 mL DI water, followed by stirring for 4 h at 400 rpm. The delaminated dispersion was washed using DI water via centrifugation at 7000 rpm until pH $\geq$ 6. Further, the sediments were re-dispersed in a small amount of DI water and the high concentration supernatant containing delaminated single- and few-layer $\text{Ti}_3\text{C}_2\text{T}_x$ MXene flakes were collected via centrifugation at 5000 rpm. The concentration of $\text{Ti}_3\text{C}_2\text{T}_x$ dispersion was measured by filtering a specific amount of colloidal solution through a polypropylene filter (3501 Coated PP, Celgard LLC, Charlotte, NC), followed by drying under a vacuum condition at 70 °C overnight and measuring the weight of the solid residue.

Synthesis of TMAOH Delaminated $\text{Ti}_3\text{C}_2\text{T}_x$ Suspensions: The etching process was the same as LiCl delaminated $\text{Ti}_3\text{C}_2\text{T}_x$. The delamination was performed by mixing 1 g of multilayered powder in 20 mL of deionized water. 1 mL of TMAOH (Sigma Aldrich, 25 wt% in water) was added to the solution. The mixture was stirred at 200 rpm for 12 h. After the reaction, the solution was washed to neutral pH by repeated centrifugation (5000 rpm, 30 min) and the sediment was redispersed with 50 mL deionized water. The solution was bath sonicated for 1 h and then centrifuged at 3500 rpm for 1 h.

Synthesis of TMAOH Delaminated Mo_2TiC_2 Suspensions: To synthesize Mo_2TiC_2 , 1 g of $\text{Mo}_2\text{TiAlC}_2$ was added to 10 mL 48–50 wt% HF and stirred for 48 h at 55 °C. Intercalation was performed by mixing 1 g of multilayered powder in 20 mL of deionized water. 1 mL of TMAOH (Sigma Aldrich, 25 wt% in water) was added to the solution. The mixture was stirred at 200 rpm for 12 h. After the reaction, the solution was washed to neutral pH by repeated centrifugation (5000 rpm, 30 min) and the sediment was redispersed with 50 mL deionized water. The solution was bath sonicated for 1 h and then centrifuged at 3500 rpm for 1 h.

Preparation of Spray-Coating and Freestanding $\text{Ti}_3\text{C}_2\text{T}_x$ Film: PET substrates were briefly cleaned by an ethanal wetted wipe and then further cleaned in a UV/ O_3 chamber. The cleaned PET substrates were then spray coated with MXene using a spray-coating gun with MXene suspension of 5 mg mL^{-1} . The spray-coating was then ended when the film turned dark green. MXene thin films were finally blow-dried and stored in the air before characterization. The freestanding $\text{Ti}_3\text{C}_2\text{T}_x$ films were prepared by vacuum filtration. The films (with a diameter of $\approx$ 4 cm) were blow-dried. The average mass loading of the films was at $\approx$ 0.8 mg cm^{-2} .

Fabrication of Patterned $\text{Ti}_3\text{C}_2\text{T}_x$ Line-Shaped Devices: The line pattern of spray-coating $\text{Ti}_3\text{C}_2\text{T}_x$ with a width of 0.5 mm and a length of 2 cm

was fabricated by a Fiber Laser Marker machine (21st Century Starlight Industrial Technology Co., Ltd), where extra Ti_3C_2 coatings aside from the line were removed by high power laser (setting power: 20%). The laser pulse (1064 nm) had a 100 ns pulse width and 20 kHz repetition rate. The laser was scanning line-by-line with a speed of 2000 mm s^{-1} . The scanning line interval was set to be 20 μm while the focused laser spot was 32 μm in diameter, which ensured that the scanned area can be fully removed by laser. The line pattern of freestanding $\text{Ti}_3\text{C}_2\text{T}_x$ films was fabricated similarly.

In Situ Resistance Test of $\text{Ti}_3\text{C}_2\text{T}_x$ Line-Shaped Devices: The patterned $\text{Ti}_3\text{C}_2\text{T}_x$ line-shaped devices were connected to a CHI 660e electrochemistry workstation. The currents for all tests were controlled to be 1 mA and the voltage variations under different conditions were tested continuously. The pulse laser which was used as the light source for optoelectronic detection was as same as the one used for laser patterning. However, to avoid rapid oxidation or laser burning of the materials, the power density of the laser was weakened by a defocusing of 2 cm. The comparison between power densities of the focused and defocused laser were displayed in Table S1 (Supporting Information). For tests in Ar, a cylinder open chamber covered by a polymethyl methacrylate plate was used with a gas inlet at the side bottom and a gas outlet at the top of the opposite side. The laser scanning area was set to be a $1 \times 1 \text{ cm}^2$ square which covered the 0.5 mm wide line-shaped device. The laser scanning direction was perpendicular to the line-shaped devices. Since the scanning line speed was 2000 mm s^{-1} and the line interval was 20 μm , it took $\approx$ 2.5 s for 1 laser scanning cycle. The average time for laser scanning on the line-shaped device was calculated to be 0.125 s.

Material Characterization: X-ray diffraction (XRD) was performed on Bruke D8 advance powder X-ray diffractometer equipped with Copper $K\alpha$ radiation ($\lambda = 1.540598 \text{ \AA}$). Raman spectra were recorded using a Horiba Raman microscope with 532 nm laser, 600 lines mm^{-1} grating at 4.02 W mm^{-2} laser power, and a 50 $\times$ objective. Spectra were acquired with a dwell time of 90 s with 2 accumulations. The morphology and microstructure of the samples were characterized by a scanning electron microscope (SEM) (Tescan Mira 3). The chemical composition of the samples was analyzed using high-resolution X-ray photoelectron spectroscopy (XPS, Thermal K-alpha+).

Calculation of the Responsivity: The responsivities of different MXene-based devices in this work are calculated and listed in Table S2 (Supporting Information). The equation for the calculation is given as follows:

$$\text{Responsivity} = \frac{I_{\text{on}} - I_{\text{off}}}{P_{\text{NIR}} \times A_{\text{MXene}}} \quad (1)$$

where I_{on} is the current of the devices under NIR laser and I_{off} represents the current without NIR laser at a given constant voltage. P_{NIR} represents the laser power density and A_{MXene} indicates the area of the laser-irradiated MXene line.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

This work was financially supported by the National Key Research and Development Project funding from the Ministry of Science and Technology of China (Grant Nos. 2021YFB3800100 and 2021YFB3800101), the National Natural Science Foundation of China (Grant No. U19A2089), the Key Fundamental Research Project funding from the Shenzhen Science and Technology Innovation Committee (Grant No. JCYJ20200109141014474), the Peacock Team Project from Shenzhen Science and Technology Innovation Committee (Grant

No. KQTD2015033110182370), the Basic and Applied Basic Research Foundation of Guangdong Province (Grant No. 2019B1515120083), and the Guangdong-Hong Kong-Macao Joint Laboratory (Grant No. 2019B121205001). X.X. would like to thank the support from Sichuan Province Science and Technology Support Program (2021JDTD0026), and the Outstanding Scholarship Fund of UESTC (A1098531023601243). C.Z. would like to acknowledge the grants from ETH Board and Empa Internal Research Call (IRC2018-CupSupercap, IRC 2020-NitFixMXene).

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

fast infrared response, infrared sensors, intercalants, MXenes, photothermal conversion

Received: March 16, 2022

Revised: April 27, 2022

Published online: June 6, 2022

- [1] D. Wang, F. Li, R. Lian, J. Xu, D. Kan, Y. Liu, G. Chen, Y. Gogotsi, Y. Wei, *ACS Nano* **2019**, 13, 11078.
- [2] M. Ghidui, M. R. Lukatskaya, M.-Q. Zhao, Y. Gogotsi, M. W. Barsoum, *Nature* **2014**, 516, 78.
- [3] C. (John) Zhang, V. Nicolosi, *Energy Storage Mater.* **2019**, 16, 102.
- [4] F. Shahzad, M. Alhabeb, C. B. Hatter, B. Anasori, S. M. Hong, C. M. Koo, Y. Gogotsi, *Science* **2016**, 353, 1137.
- [5] D. Er, J. Li, M. Naguib, Y. Gogotsi, V. B. Shenoy, *ACS Appl. Mater. Interfaces* **2014**, 6, 11173.
- [6] L. Ding, Y. Wei, Y. Wang, H. Chen, J. Caro, H. Wang, *Angew. Chem., Int. Ed.* **2017**, 56, 1825.
- [7] X. Xiao, P. Urbankowski, K. Hantanasirisakul, Y. Yang, S. Sasaki, L. Yang, C. Chen, H. Wang, L. Miao, S. H. Tolbert, S. J. L. Billinge, H. D. Abruña, S. J. May, Y. Gogotsi, *Adv. Funct. Mater.* **2019**, 29, 1809001.
- [8] G. Gao, A. P. O'Mullane, A. Du, *ACS Catal.* **2017**, 7, 494.
- [9] X. Xiao, H. Wang, W. Bao, P. Urbankowski, L. Yang, Y. Yang, K. Maleski, L. Cui, S. J. L. Billinge, G. Wang, Y. Gogotsi, *Adv. Mater.* **2019**, 31, 1902393.
- [10] Y. Xie, Y. Dall'Agnese, M. Naguib, Y. Gogotsi, M. W. Barsoum, H. L. Zhuang, P. R. C. Kent, *ACS Nano* **2014**, 8, 9606.
- [11] S. J. Kim, H.-J. Koh, C. E. Ren, O. Kwon, K. Maleski, S.-Y. Cho, B. Anasori, C.-K. Kim, Y.-K. Choi, J. Kim, Y. Gogotsi, H.-T. Jung, *ACS Nano* **2018**, 12, 986.
- [12] X. Xiao, H. Wang, P. Urbankowski, Y. Gogotsi, *Chem. Soc. Rev.* **2018**, 47, 8744.
- [13] R. Zeng, W. Wang, M. Chen, Q. Wan, C. Wang, D. Knopp, D. Tang, *Nano Energy* **2021**, 82, 105711.
- [14] G. Cai, Z. Yu, P. Tong, D. Tang, *Nanoscale* **2019**, 11, 15659.
- [15] J. Chen, P. Tong, L. Huang, Z. Yu, D. Tang, *Electrochim. Acta* **2019**, 319, 375.
- [16] L. Lu, X. Han, J. Lin, Y. Zhang, M. Qiu, Y. Chen, M. Li, D. Tang, *Analyst* **2021**, 146, 2664.
- [17] Y. Wu, W. Xu, W. Tang, Z. Wang, Y. Wang, Z. Lv, Y. Zhang, W. Zhong, H.-L. Cai, R. Yang, X. S. Wu, *Nano Energy* **2021**, 90, 106532.
- [18] Y. Wu, D. Yang, Y. Zhang, S. Jiao, W. Tang, Z. Wang, N. Wu, Y. Wang, W. Zhong, A. Zhang, J. Hao, H.-L. Cai, X. S. Wu, *Chem. Eng. J.* **2022**, 439, 135640.
- [19] Y. Wu, W. Xiong, Z. Wang, Y. Wang, K. Sun, X. Song, Z. Lv, W. Xu, W. Zhong, X. Zou, H.-L. Cai, X. Wu, *Chem. Eng. J.* **2022**, 427, 131925.
- [20] Y. Wu, X. Song, W. Xu, K. Sun, Z. Wang, Z. Lv, Y. Wang, Y. Wang, W. Zhong, J. Wei, H.-L. Cai, X. Wu, *Small* **2021**, 17, 2101705.
- [21] M. Naguib, M. Kurtoglu, V. Presser, J. Lu, J. Niu, M. Heon, L. Hultman, Y. Gogotsi, M. W. Barsoum, *Adv. Mater.* **2011**, 23, 4248.
- [22] J. Zhang, N. Kong, S. Uzun, A. Levitt, S. Seyedin, P. A. Lynch, S. Qin, M. Han, W. Yang, J. Liu, X. Wang, Y. Gogotsi, J. M. Razal, *Adv. Mater.* **2020**, 32, 2001093.
- [23] V. Kamysbayev, A. S. Filatov, H. Hu, X. Rui, F. Lagunas, D. Wang, R. F. Klie, D. V. Talapin, *Science* **2020**, 369, 979.
- [24] B. Anasori, M. R. Lukatskaya, Y. Gogotsi, *Nat. Rev. Mater.* **2017**, 2, 16098.
- [25] X. Wang, M. Salari, D. Jiang, J. C. Varela, B. Anasori, D. J. Wesolowski, S. Dai, M. W. Grinstaff, Y. Gogotsi, *Nat. Rev. Mater.* **2020**, 5, 787.
- [26] X. Wang, T. S. Mathis, K. Li, Z. Lin, L. Vlcek, T. Torita, N. C. Osti, C. Hatter, P. Urbankowski, A. Sarycheva, M. Tyagi, E. Mamontov, P. Simon, Y. Gogotsi, *Nat. Energy* **2019**, 4, 241.
- [27] H. Xu, A. Ren, J. Wu, Z. Wang, *Adv. Funct. Mater.* **2020**, 30, 2000907.
- [28] M. Han, C. E. Shuck, R. Rakhmanov, D. Parchment, B. Anasori, C. M. Koo, G. Friedman, Y. Gogotsi, *ACS Nano* **2020**, 14, 5008.
- [29] D. B. Velusamy, J. K. El-Demellawi, A. M. El-Zohry, A. Giugni, S. Lopatin, M. N. Hedhili, A. E. Mansour, E. D. Fabrizio, O. F. Mohammed, H. N. Alshareef, *Adv. Mater.* **2019**, 31, 1807658.
- [30] Z. Liu, H. N. Alshareef, *Adv. Electron. Mater.* **2021**, 7, 2100295.
- [31] S. Chertopalov, V. N. Mochalin, *ACS Nano* **2018**, 12, 6109.
- [32] J. Chen, Z. Li, F. Ni, W. Ouyang, X. Fang, *Mater. Horiz.* **2020**, 7, 1828.
- [33] K. Montazeri, M. Currie, L. Verger, P. Dianat, M. W. Barsoum, B. Nabet, *Adv. Mater.* **2019**, 31, 1903271.
- [34] S. Hong, G. Zou, H. Kim, D. Huang, P. Wang, H. N. Alshareef, *ACS Nano* **2020**, 14, 9042.
- [35] M. Long, P. Wang, H. Fang, W. Hu, *Adv. Funct. Mater.* **2019**, 29, 1803807.
- [36] J. Cheng, C. Wang, X. Zou, L. Liao, *Adv. Opt. Mater.* **2019**, 7, 1800441.
- [37] B. Li, Q.-B. Zhu, C. Cui, C. Liu, Z.-H. Wang, S. Feng, Y. Sun, H.-L. Zhu, X. Su, Y.-M. Zhao, H.-W. Zhang, J. Yao, S. Qiu, Q.-W. Li, X.-M. Wang, X.-H. Wang, H.-M. Cheng, D.-M. Sun, *Adv. Mater.* **2022**, 34, 2201298.
- [38] J. L. Hart, K. Hantanasirisakul, A. C. Lang, B. Anasori, D. Pinto, Y. Pivak, J. T. van Ommen, S. J. May, Y. Gogotsi, M. L. Taheri, *Nat. Commun.* **2019**, 10, 522.
- [39] D. Xu, Z. Li, L. Li, J. Wang, *Adv. Funct. Mater.* **2020**, 30, 2000712.
- [40] X. Han, J. Huang, H. Lin, Z. Wang, P. Li, Y. Chen, *Adv. Healthcare Mater.* **2018**, 7, 1701394.
- [41] S. Zada, W. Dai, Z. Kai, H. Lu, X. Meng, Y. Zhang, Y. Cheng, F. Yan, P. Fu, X. Zhang, H. Dong, *Angew. Chem.* **2020**, 132, 6663.
- [42] S. Xu, X. Bai, L. Wang, *Inorg. Chem. Front.* **2018**, 5, 751.
- [43] X. Yu, X. Cai, H. Cui, S.-W. Lee, X.-F. Yu, B. Liu, *Nanoscale* **2017**, 9, 17859.
- [44] H. Cheng, Q. Liu, S. Han, S. Zhang, X. Ouyang, X. Wang, Z. Duan, H. Wei, X. Zhang, N. Ma, M. Xue, *ACS Appl. Mater. Interfaces* **2020**, 12, 37637.
- [45] R. Li, L. Zhang, L. Shi, P. Wang, *ACS Nano* **2017**, 11, 3752.
- [46] H. Deng, J. Liao, J. Piao, Y. Zhang, S. He, Q. Zhou, Y. Li, L. Zhang, *J. Appl. Polym. Sci.* **2020**, 137, 49137.
- [47] H. Lin, X. Wang, L. Yu, Y. Chen, J. Shi, *Nano Lett.* **2017**, 17, 384.
- [48] A. Sarycheva, A. Polemi, Y. Liu, K. Dandekar, B. Anasori, Y. Gogotsi, *Sci. Adv.* **2018**, 4, 0920.